
A
Same-Vial Viral RNA Doubles Host Transcriptome Discovery
Significant Human Genes at BH-FDR q < 0.05
663
(3.3% | π₁=27%)
6,087
(30.2% | π₁=50%)
9,819
(48.7% | π₁=65%)
10,580
(52.5% | π₁=68%)
12,863
(63.8% | π₁=76%)
13,800
(68.5% | π₁=78%)
Exogenous IR1 (WGS DNA)
2019 WGS log2(EBV CN)
Spliced BZLF1 (Zta) RPM
BGLF5 (Host-Shutoff)
log2(Late / Latent mRNA)
LMP-1 RPKM (NF-kB Driver)
B
Baron-Kenny Mediation Asymmetry: Spliced BZLF1 Mediates CN
WGS CN | Spliced BZLF1 (92.4% of 6,087 hits eliminated!)
Spliced BZLF1 | WGS CN (78.7% of 9,819 hits survive!)
No attenuation (y = x)
Forward slope = 0.20 (86.0% of R² mediated)
Reverse slope = 0.74 (direct viral RNA effect)
Conditioned Baron-Kenny Partial Correlation (r)
Unconditioned Within-Pop+Batch Partial Correlation (r)
C
Exogenous DNA Instrument (IR1) Reconstructs Host Signature
20,154-gene alignment: r = +0.7636 (663 IR1 genes q < 0.05)
CBARP
USP17L2
LINC01443
HLA-DOA
HLA-DOB
CD40
EBF1
WGS log2(EBV CN) Effect on Host Gene (r)
Exogenous Viral DNA IR1 Repeat Effect on Host Gene (r)
D
Bipartite 4-Axis Viral-to-Host Functional Module Map
IFIH1 (NF-kB) +0.76 -0.32 -0.34 +0.45 -0.34
TNFAIP3 (NF-kB) +0.74 -0.31 -0.32 +0.40 -0.25
CD40 (B-Cell) +0.67 -0.40 -0.36 +0.37 -0.35
ICAM1 (NF-kB) +0.64 -0.23 -0.24 +0.35 -0.20
NFKBIA (NF-kB) +0.63 -0.29 -0.31 +0.32 -0.12
EBF1 (B-Cell) +0.46 -0.50 -0.48 +0.22 -0.19
MYC (B-Cell) +0.41 -0.29 -0.19 +0.10 -0.03
TNFRSF17 (Plasmablast) -0.62 +0.35 +0.30 -0.43 +0.18
MZB1 (Plasmablast) -0.51 +0.23 +0.18 -0.31 +0.13
SSR4 (Plasmablast) -0.51 +0.19 +0.16 -0.39 +0.10
JCHAIN (Plasmablast) -0.36 +0.25 +0.14 -0.08 -0.02
MX1 (IFN / ISG) +0.28 -0.06 -0.15 +0.42 -0.23
IFIT1 (IFN / ISG) +0.29 +0.03 -0.06 +0.38 -0.32
ISG15 (IFN / ISG) +0.12 +0.13 +0.00 +0.37 -0.24
OAS1 (IFN / ISG) +0.37 -0.03 -0.08 +0.28 -0.28
STAT1 (IFN / ISG) +0.24 +0.04 -0.06 +0.23 -0.23
LMP-1 (Latent NF-kB)
BGLF5 (Lytic / Shutoff)
Spliced BZLF1 (Lytic Switch)
EBNA-2 (Latent III)
BART lncRNA (ISG Repressor)
Within-Pop+Batch Partial r